

WASP-10b

R-1752

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_130918 · created 2026-08-02T20:02:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456553.6949 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.088 +/- 0.061
Transit depth (Rp/Rs)^2	0.77 +/- 1.08 [%]
Orbital Inclination (inc)	84.45 +/- 1.9
Airmass coefficient 1 (a1)	2.31 +/- 1.39
Airmass coefficient 2 (a2)	-0.74 +/- 0.99
Scatter in the residuals of the lightcurve fit is	65.3 %
Best Comparison Star	#3 - [157, 29]
Optimal Aperture	1.24
Optimal Annulus	10.35
Transit Duration (day)	0.029 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.088 $\pm$ 0.061 vs 0.1592 (deviation 0.9 σ)
Duration fit vs published	0.029 d vs 0.0946 d (deviation 1.58 σ)
Mid-time O-C	-0.2 min
Depth SNR	1.1
Residual scatter	65.3 %

Light curve

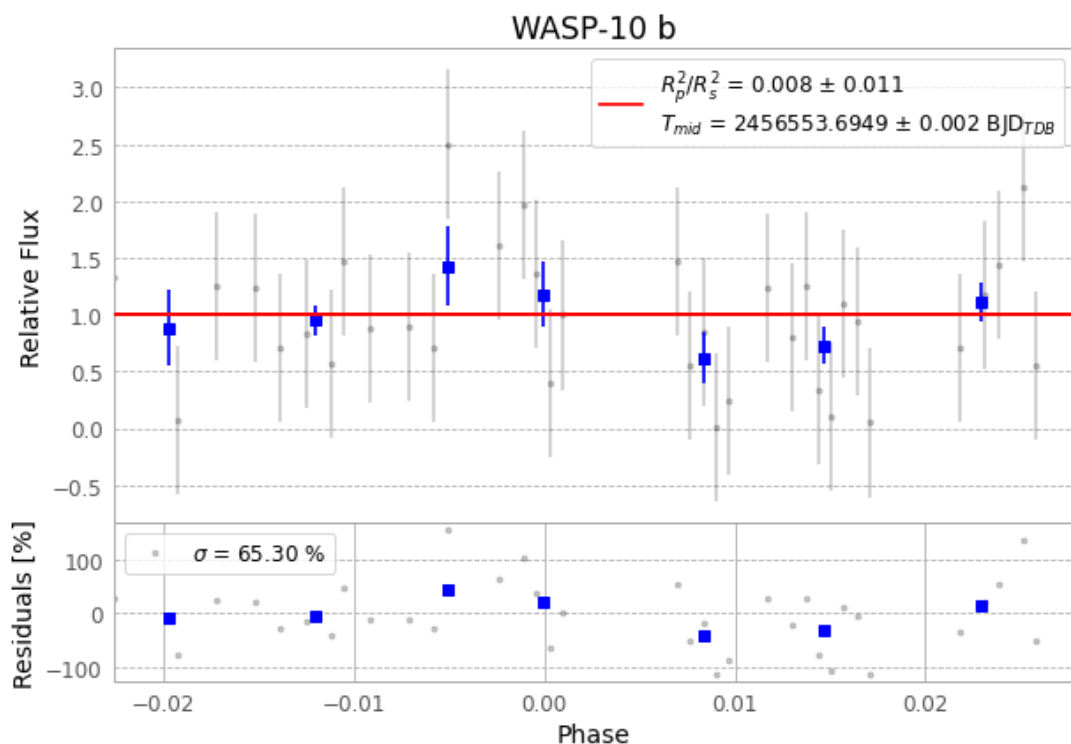


Figure 1 — FinalLightCurve_WASP-10 b_2013-09-17.png (SHA-256 7309ef7952559fa2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [235, 228], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8bd1bdf294831424...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-10130918024812.FITS → WASP-10130918063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-130918.log (2a7d32bcbe8f...), FinalLightCurve_WASP-10 b_2013-09-17.png (7309ef795255...), FinalLightCurve_WASP-10 b_2013-09-17.csv (b0b8832c07f3...)

Compute resources

Wall time 250.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 20:03Z.